

The 2-phase method

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Consider a linear program (LP) in standard form:

$$\text{Minimize } Z_P = c^T x$$

Subject to

$$Ax = b \quad (LP)$$

$$x \geq 0_n,$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$. We assume $\text{rank}(A) = m$. The set $N := \{1, \dots, n\}$ is used to index the variables/columns of A . We may assume $b \geq 0_m$, since the equalities $Ax = b$ can be scaled with (-1) without changing the set of solutions.

A *basis* of (LP) is an m -subset $B \subseteq N$ of linearly independent columns of A . The basic solution corresponding to B is the unique solution $\bar{x}$ to the system:

$$Ax = b \quad \text{and} \quad x_j = 0 \quad \text{for all } j \in N \setminus B$$

A basis B is called *feasible* for (LP) if the corresponding basic solution $\bar{x}$ is a feasible solution to (LP).

The (primal) simplex method can solve (LP) starting from a feasible basis $B \subseteq N$. However, it is not clear how one obtains a feasible basis. The 2-phase method solves (LP) by first finding a feasible basis to (LP) in phase 1, and then applying the "usual" simplex method in phase 2. In this note we only describe phase 1. Phase 2 has been described in earlier notes.

In phase 1 the following "artificial" linear optimization problem is solved with the usual simplex method:

$$\text{Minimize } \sum_{i=1}^m y_i$$

Subject to

$$Ax + y = b \quad (LP^a)$$

$$x \geq 0_n \text{ and } y \geq 0_m,$$

Observe that the feasible bases for (LP) are exactly the optimal bases for (LP^a) that do not contain any y -variables. Also, the coefficient matrix for (LP^a) is $[A, I_m]$, so the last m columns of $[A, I_m]$ forms a feasible basis for (LP^a) (since we have assumed $b \geq 0_m$). It follows that (LP^a) can be solved with the (primal) simplex method starting from a simple feasible basis.

We now describe phase 1 on an example. The method for a general linear optimization problem should be clear from this example. Consider the following linear optimization problem:

$$\begin{aligned} &\text{Minimize } Z_P = 3x_1 + 5x_2 \\ &\text{Subject to} \\ &\quad 3x_1 + 2x_2 + 5x_3 \geq 4 \\ &\quad 2x_1 + x_2 + x_3 \geq 3 \\ &\quad x_1 + 5x_2 + x_3 \leq 5 \\ &\quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

Adding slack/surplus variables gives a linear optimization problem (LP) in standard form:

$$\begin{aligned} &\text{Minimize } Z_P = 3x_1 + 5x_2 \\ &\text{Subject to} \\ &\quad 3x_1 + 2x_2 + 5x_3 - x_4 = 4 \\ &\quad 2x_1 + x_2 + x_3 - x_5 = 3 \\ &\quad x_1 + 5x_2 + x_3 + x_6 = 5 \\ &\quad x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{aligned}$$

In this example one has $N = \{1, \dots, 6\}$. We now describe how to obtain a feasible basis $B \subseteq N$ for (LP) with the 2-phase method. Phase 1 solves the artificial problem (LP^a) . To construct (LP^a) , non-negative artificial variables y_1, y_2 and y_3 are added, and the objective is to minimize $y_1 + y_2 + y_3$:

$$\begin{aligned} &\text{Minimize } y_1 + y_2 + y_3 \\ &\text{Subject to} \\ &\quad 3x_1 + 2x_2 + 5x_3 - x_4 + y_1 = 4 \\ &\quad 2x_1 + x_2 + x_3 - x_5 + y_2 = 3 \\ &\quad x_1 + 5x_2 + x_3 + x_6 + y_3 = 5 \\ &\quad x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{aligned}$$

or written in tableau form:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	y_1	y_2	y_3	Right hand side
Z_P	-1	0	0	0	0	0	0	1	1	1	0
y_1	0	3	2	5	-1	0	0	1	0	0	4
y_2	0	2	1	1	0	-1	0	0	1	0	3
y_3	0	1	5	1	0	0	1	0	0	1	5

The (primal) simplex method is now applied to solve (LP^a) . Firstly the objective row of the tableau must have zeros on entries corresponding to basic variables, *i.e.*, the columns in the tableau corresponding to the basic variables y_1 , y_2 and y_3 must be unit vectors. To achieve this, one subtract the last 3 rows of the tableau from the first row of the tableau:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	y_1	y_2	y_3	Right hand side
Z_P	-1	-6	-8	-7	1	1	-1	0	0	0	-12
y_1	0	3	2	5	-1	0	0	1	0	0	4
y_2	0	2	1	1	0	-1	0	0	1	0	3
y_3	0	1	5	1	0	0	1	0	0	1	5

This is now the tableau corresponding to the start basis y_1 , y_2 and y_3 for (LP^a) . Next we choose x_6 to enter the basis (since it has a negative reduced cost). The ratio computation gives that y_3 must leave the basis. Row operations are then performed to obtain the next simplex tableau. Observe that once an artificial variable leaves the basis one never wants that variable to enter the basis again. Hence the column corresponding to y_3 is deleted from the remaining tableaus. The next tableau is therefore:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	y_1	y_2	Right hand side
Z_P	-1	-5	-3	-6	1	1	0	0	0	-12
y_1	0	3	2	5	-1	0	0	1	0	4
y_2	0	2	1	1	0	-1	0	0	1	3
x_6	0	1	5	1	0	0	1	0	0	5

There are still variables with negative reduced costs and artificial variables in the basis. The simplex method therefore continues. We choose x_1

to enter the basis. The ratio computation shows that this implies that y_1 leaves the basis. Performing row operations on the tableau gives the next tableau:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	y_2	Right hand side
Z_P	-1	0	$\frac{1}{3}$	$\frac{7}{3}$	$-\frac{2}{3}$	1	0	0	$-\frac{1}{3}$
x_1	0	1	$\frac{2}{3}$	$\frac{5}{3}$	$-\frac{1}{3}$	0	0	0	$\frac{4}{3}$
y_2	0	0	$-\frac{1}{3}$	$-\frac{7}{3}$	$\frac{2}{3}$	-1	0	1	$\frac{1}{3}$
x_6	0	0	$\frac{13}{3}$	$-\frac{2}{3}$	$\frac{1}{3}$	0	1	0	$\frac{11}{3}$

There is now only one variable with negative reduced cost, *i.e.*, x_4 is the only candidate for an entering non-basic variable. The ratio computation gives that y_2 must leave the basis. This results in the final tableau, where there are no basic artificial variables:

Basic Variable	Z_P	x_1	x_2	x_3	x_4	x_5	x_6	Right hand side
Z_P	-1	0	0	0	0	0	0	0
x_1	0	1	$\frac{1}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	0	$\frac{3}{2}$
x_4	0	0	$-\frac{1}{2}$	$-\frac{7}{2}$	1	$-\frac{3}{2}$	0	$\frac{1}{2}$
x_6	0	0	$\frac{9}{2}$	$\frac{1}{2}$	0	$\frac{1}{2}$	1	$\frac{7}{2}$

We conclude that the final basis $B = \{1, 4, 6\}$ is a feasible basis for both the artificial problem (LP^a) and the original problem (LP).